

SATO CL4NX Plus RFID
Printer

The SATO CL4NX Plus is a best-in-class 4-inch industrial thermal label printer featuring a die-cast aluminum frame, dual CPU architecture, up to 14 ips print speed at 203/305 dpi, and an optional fully integrated UHF or HF/NFC RFID encoder module. Designed for demanding production environments, it supports a 254 mm (10 in) media roll, 600 m ribbon capacity, tool-free maintenance, and a comprehensive suite of emulation languages.



PRINT SPECIFICATIONS

Print Method	Direct Thermal / Thermal Transfer
Resolution	8 dots/mm (203 dpi), 12 dots/mm (305 dpi), 24 dots/mm (609 dpi)
Print Speed	203/305 dpi: 2–14 ips (50–355 mm/sec); 609 dpi: 2–6 ips (50–152 mm/sec)
Print Width	104 mm (4.09 in)
Print Length	Max. 2500 mm (98.42 in)
Print Mode	Continuous, Tear-off, Cutter, Dispenser, Peel & Present, Linerless
Display	3.5-inch full-color LCD

RFID SPECIFICATIONS

Air Interface Protocol	EPC C1Gen2 (ISO/IEC 18000-6C) [UHF]; ISO/IEC 15693 & ISO/IEC 14443 Type A [HF/NFC]
UHF Frequency	ETSI: 865–868 MHz; FCC: 902–928 MHz
RFID Module	Fully Integrated UHF RFID Reader/Encoder Module
SATO RF Analyze (SRA)	Automatic antenna and inlay setting optimization
Metal Tag Support	Compatible with flexible metal RFID tags (Confidex, Epozzy)

MEDIA SPECIFICATIONS

Media Type	Roll (Continuous, Die-cut), Fan-fold, Tag, Ticket
Label Width	Min. 22 mm (0.87 in) – Max. 128 mm (5.04 in)
Label Length	Min. 6 mm (0.24 in) – Max. 2500 mm (98.42 in)
Media Roll OD	Max. 254 mm (10.0 in)
Media Core ID	76 mm (3.0 in)
Media Thickness	0.06 – 0.25 mm (2.4 – 10.0 mil)
Ribbon Width	39.5 mm (1.55 in) – 128 mm (5.04 in)
Ribbon Length	Max. 600 m (1968.5 ft)
Ribbon Core ID	25.4 mm (1.0 in)

CONNECTIVITY

Data Interface (Standard)	USB 2.0 (Type A & Type B, Front & Rear), Ethernet (IPv4/IPv6), RS-232C, Parallel (IEEE 1284), EXT I/O Interface, NFC
Data Interface (Optional)	Wi-Fi & Wi-Fi Direct (IEEE 802.11 a/b/g/n, Dual Band 2.4 GHz / 5 GHz), Bluetooth 3.0

SYSTEM

CPU	Dual CPU / Dual OS; CPU1: 800 MHz (Linux); CPU2: 800 MHz (ITRON OS)
Memory	CPU1: 2 GB Flash, 256 MB RAM; CPU2: 4 MB Flash, 64 MB RAM
Emulation (Auto Detect)	SBPL, SEPL, SZPL, SDPL, SIPL, STCL
Additional	XML, Direct PDF Printing, AEP (Application Enabled Printing)
Label Sensor	Adjustable Transmissive Sensor (Gap), Adjustable Reflective Sensor (Black Mark)

PHYSICAL SPECIFICATIONS

Dimensions (W x D x H)	271 x 457 x 321 mm (10.7 x 18.0 x 12.6 in)
Weight	Approx. 15.1 kg (33.3 lbs)
Construction	Die-cast aluminum frame with metal casing

ELECTRICAL SPECIFICATIONS

Power Supply	Universal Auto-ranging; AC 100–240 V ±10%, 50/60 Hz
Power Consumption (Peak)	190 VA / 180 W (at 30% printing ratio)
Power Consumption (Standby)	40 VA / 20 W

ENVIRONMENTAL SPECIFICATIONS

Operating Temperature	0 – 40 °C (32 – 104 °F)
Operating Humidity	30 – 80% RH (non-condensing)
Storage Temperature	-20 – 60 °C (-4 – 140 °F)
Storage Humidity	30 – 90% RH (non-condensing)

FEATURE HIGHLIGHTS

- Best-in-class print speed: up to 14 ips at 203/305 dpi
- Available in 203, 305, and 609 dpi resolutions
- Optional fully integrated UHF RFID (EPC C1Gen2 / ISO/IEC 18000-6C) or HF/NFC RFID encoder
- Compatible with flexible metal RFID tags (Confidex, Epozzy)
- Snap-on print head with tool-free platen changing
- 3.5-inch full-color LCD with on-board instructional videos
- Dual CPU / Dual OS architecture (Linux + ITRON OS)
- High-capacity: 254 mm (10 in) media roll OD, 600 m ribbon
- AEP (Application Enabled Printing) for on-printer custom applications
- Supports 47 print languages and 31 display languages

PRODUCT GROUP

Category	Industrial RFID Printer
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